Water Usage in Households: An Exploratory Data Analysis

Evgenie Tivaniuk

September 2025

1 Introduction

This report presents an exploratory data analysis of the "Smart Water Monitoring Systems" dataset.

The study focuses on household water consumption and its relationship with factors such as housing quality, income level, number of residents, and number of guests.

For a deeper investigation, three research hypotheses were formulated and tested using statistical methods. Each hypothesis was evaluated and either confirmed or disproved based on the results of the analysis.

All plots were produced in Python using libraries such as Seaborn and Matplotlib; statistics include Pearson correlations and Chi-square tests.

The data set is publicly available at: https://www.hackerearth.com/problem/machine-learning/smart-water-monitoring-systems-8-c169bb31/

Analyzed file: `water_train.csv`

2 Research Questions

This study is guided by three research hypotheses that aim to examine the relationship between household water consumption and socioeconomic as well as demographic factors:

1. For households with a low income level, water consumption per person does not decrease as water price increases.
2. Guests have a lower impact on total Water Consumption (not per person) compared to Residents.
3. Residents can own only specific Apartment Types that fit their Income Level and not exceed it.

3 Descriptive Analysis

The dataset has 14,000 entries consisting of various input features of the recorded population such as the number of residents, apartment type, water price, income level, number of guests, outdoor amenities, temperature and humidity. The output feature is Water Consumption. Other features of the provided dataset will not be examined or discussed in this research.

The dataset has missing/corrupted data, and the 'Apartment Type' feature is poorly designed, mixing flats and houses. This complicates comparison of housing luxury levels, so feature engineering was applied (see Hypothesis 3). Exact region of where the data was collected is unknown. While we assume that the data was gathered from India, we can't validate that.

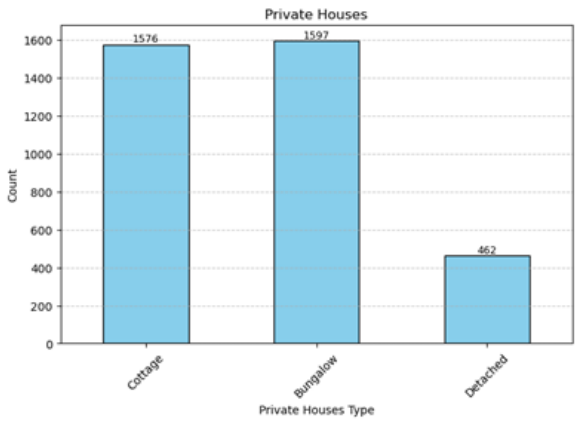
Table 1: Key Descriptive Characteristics of the Dataset

Feature	**Values / Description**
Dwellings	Two main types: Private houses and Flats.
Private houses	Subtypes: Cottage, Bungalow (most common), Detached.
Flats	Subtypes: Studio, 1BHK, 2BHK (most common), 3BHK.
Residents number	Range: 1–5 residents; most common: 3.
Income level	Categories: Low, Middle (most common), Upper Middle, Rich.
Water consumption	Min: 35.54 Max: 531.49 Mean: 164.46 Median: 150.37

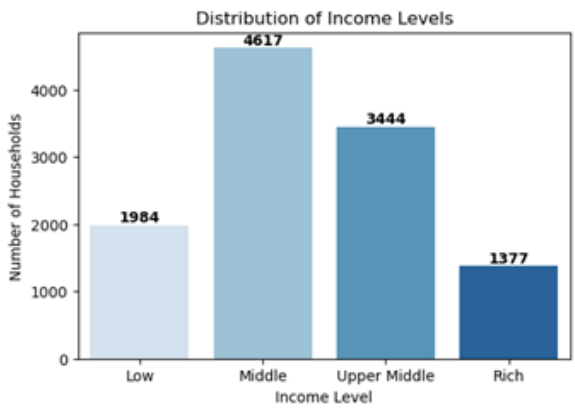
Note: 1BHK = one-bedroom flat, 2BHK = two-bedroom flat, 3BHK = three-bedroom flat.

Correlation analysis shows negligible effect of Temperature and Humidity on water consumption, but a strong relation between Income Level and Apartment Type (see Hypothesis 3). This link is important for potential prediction models and will require deeper feature engineering.

For private houses, the data distribution by Apartment Type is the following.

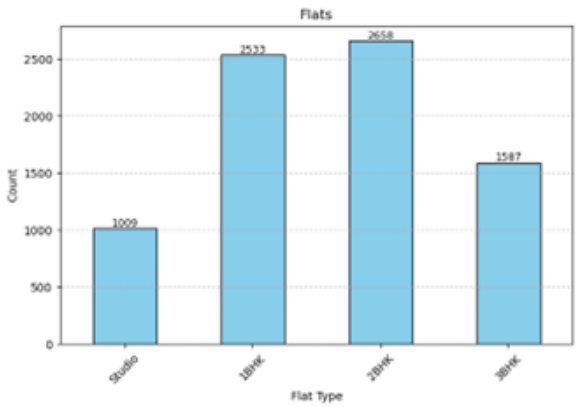


Most private-house residents live in Cottages or Bungalows, while Detached houses are rare. As the dataset lacks details on room count or luxury, this distribution helps approximate housing luxury levels.



Income levels are near-normal: most households are Middle or Upper-Middle, with far fewer Low or Rich.

For flats in multi-story buildings, the data distribution by Apartment Type is the following.



Most flat residents live in 1- or 2-room apartments; studios are typical for poorer households, while 3-room flats are usually reserved for richer ones.

Table 2: Missing and Invalid Values

Column	**Issue type**	**Count**
Residents	Negative values	280
Guests	Negative values	153
Apartment_Type	Missing values	426
Income_Level	Missing values	426
Income_Level	Invalid category values	416
Temperature	Missing values	441
Humidity	Non-numeric or missing values	397
Appliance_Usage	Missing values	415

4 Advanced Analysis

4.1 Hypothesis 1: For households with a low income level, water consumption per person does not decrease as water price increases.

Water is a basic necessity, so usage cannot fall below a minimum, regardless of price or income. **Hypothesis:** For low-income households, per-person water consumption does not decrease as water price increases.

Table 3: Summary statistics: water consumption per person

	Low income	Non-low
Q1	33.05	37.57
Median	41.04	45.91
Mean	43.52	47.67
Q3	50.85	55.62
IQR	17.81	18.04
Outliers %	2.37	1.60

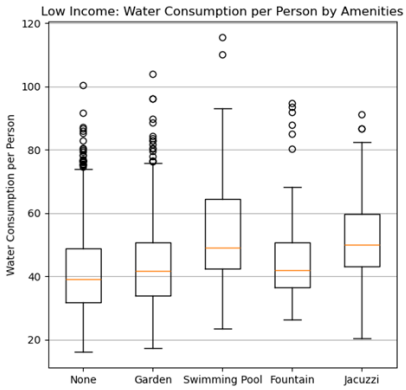


Figure 1: Amenities: Low income

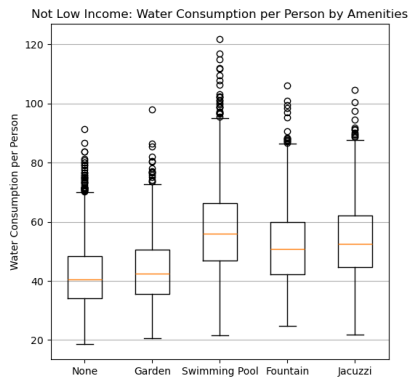


Figure 2: Amenities: Non-low

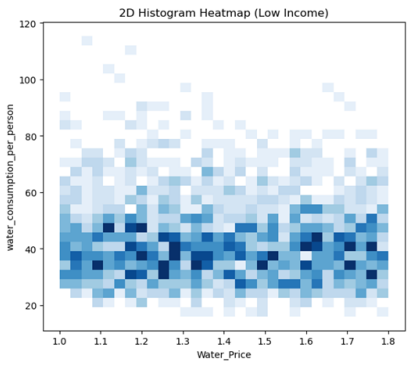


Figure 3: Heatmap: Low income

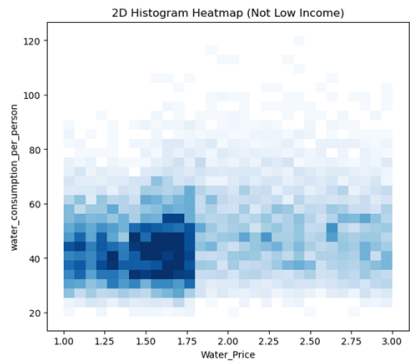


Figure 4: Heatmap: Non-low

Conclusions.

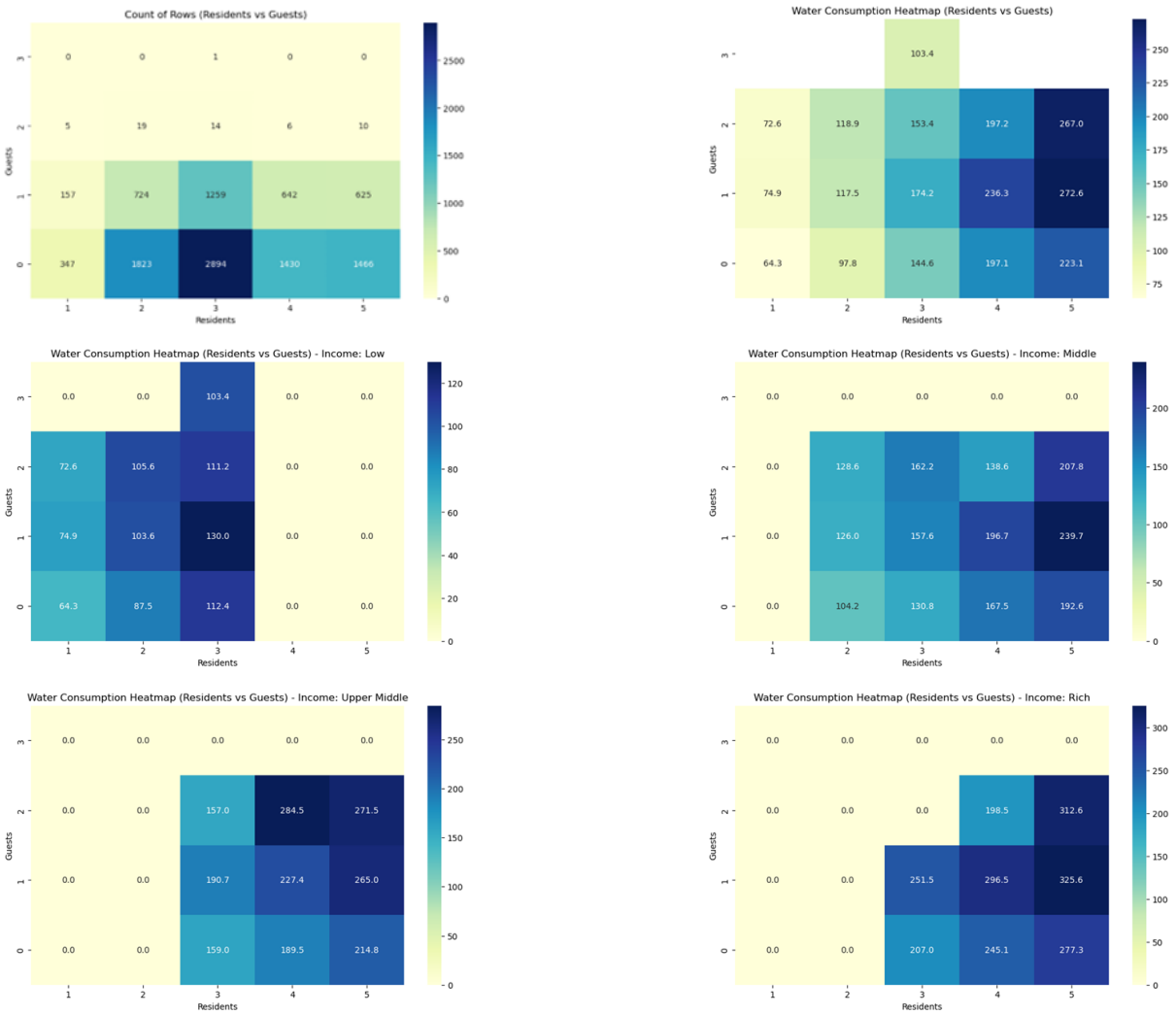
- Q1 values are nearly identical across income groups, confirming a stable baseline.
- Gardens for low-income likely represent basic yards/patches, which is a necessity.
- Pools and fountains increase consumption but are luxury items, so they do not contradict the baseline.
- Heatmaps confirm water consumption stability across different water prices.

Hypothesis 1 confirmed: minimum water use remains constant across income levels and prices.

4.2 Hypothesis 2: Guests have a lower impact on total Water Consumption (not per person) compared to Residents

Guests are temporary users (limited activities), while residents drive most daily water use. **Hypothesis:** Guests have a smaller impact on total household water consumption than residents.

Figure 5: Water Consumption Heatmaps (Residents vs Guests)



Conclusions.

- It cannot be stated with certainty that guests always have a lower impact than residents.
- Data for households with multiple guests is very limited, which reduces reliability.
- Single resident–guest cases are overrepresented; they bias the analysis and cannot be generalized.
- Most other resident–guest–income combinations indicate the hypothesis does not hold.
- Residents are highly correlated with Income Level (see Hypothesis 3).
- Formal tests (e.g., two-way ANOVA) were not feasible due to sparse data; we relied on descriptive patterns and heatmaps.

Hypothesis 2 disproved: guests do not consistently show a lower impact on water consumption than residents.

4.3 Hypothesis 3: Residents can own only specific Apartment Types that fit their Income Level and not exceed it

By economic reasoning, higher-income households should own higher-luxury dwellings. **Hypothesis:** residents can only own apartment types that fit their income level and not exceed it. Inheritance may explain some mismatches, but the scale suggested by the raw data was implausible. Another issue: the original feature *Apartment Type* mixed Flats and Private Houses, leading to false results.

Initial observation (incorrect). Original distributions suggested low- and middle-income households sometimes owned "luxury" dwellings (e.g., Cottages, Bungalows), seemingly disproving the hypothesis. This was later shown to be a design flaw in the data.

Feature engineering. We redefined the feature:

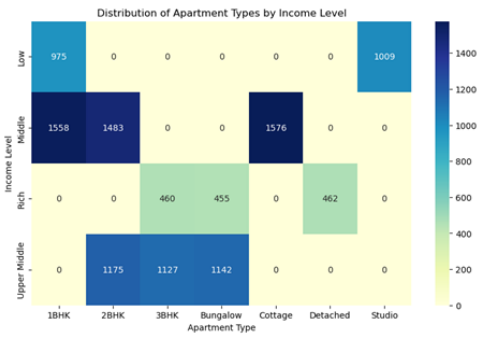
- Separated dwellings into Flats and Private Houses, and created separate quality scales for each.
- Luxury scales: Flats: Studio < 1BHK < 2BHK < 3BHK; PriHouses: Cottage < Bungalow < Detached.
- Correlation with Income Level increased: original 0.72 → Flat quality 0.84, House quality 0.91.

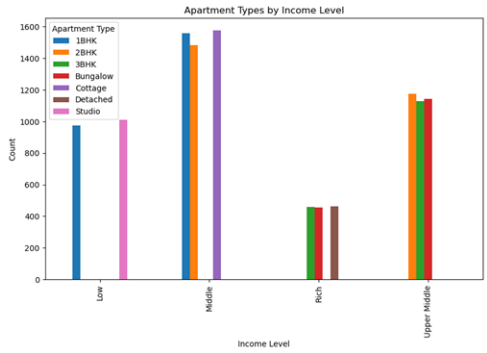
Distributions and Chi-square Test Validation

Table 4: Chi-square test: Income Level × Apartment Type

Income Level	1BHK	2BHK	3BHK	Bungalow	Cottage	Detached	Studio
Low	975	0	0	0	0	0	1009
Middle	1558	1483	0	0	1576	0	0
Upper Middle	0	1175	1127	1142	0	0	0
Rich	0	0	460	455	0	462	0
Chi-square: 17,962.8 **df:** 18 **p-value:** < 0.001 **Cramér's V:** 0.724							

Note: Association between Income Level and Apartment Type is statistically significant. Effect size indicates a **strong** relationship (Cramér's V ≈ 0.72).





Conclusions.

- The raw Apartment Type feature was misleading and required feature engineering.
- Separating Flats and Houses clarified the luxury scale and improved correlations.
- New distributions show each income group restricted to specific types, neither higher nor lower than expected.
- Chi-square results confirm a strong statistical association.

Hypothesis 3 Confirmed: each income group can only own apartment types consistent with its expected luxury level, neither higher nor lower than expected.

5 Conclusions and Insights

Hypotheses

- Hypothesis 1: Confirmed;
- Hypothesis 2: Disproved;
- Hypothesis 3: Confirmed (after feature engineering);

Insights

- Low-income "garden" often = small yard/patch, not luxury.
- Guest/resident distribution is imbalanced: very few $\geq$2 guests; many cases of (1 Guest + 1 Resident) cases bias results.
- Residents correlate with income (1→Low; 2→Low/Mid; 4→Mid/Upper/Rich).
- Strong link: Income Level with Apartment Type (χ^2, Cramér's $V \approx 0.72$).
- Original *Apartment Type* is misleading; must be engineered (Flats vs Houses).
- New features (Flat/House) correlate even stronger ⇒ further engineering needed.
- Future research: in which cases low-income households might own non-garden amenities.

6 Discussion

Two hypotheses confirmed, one disproved; dataset flaws and extra insights identified.

Main points.

- Water demand has a stable baseline (necessity).
- Guests' effect not weaker or unstable than residents'.
- Income Level strongly tied to Apartment Type.
- Dataset flaws: unclear geography (based on dataset online source we assume the data is from India); Apartment Type mixed flats/houses; Amenities partly unreliable; guest distribution highly imbalanced.
- Future actions: clarify geography, gather more data for cases of $\geq$2 guests, add Flat/House flag, validate Amenities.

Creative insights.

- Original *Apartment Type* is misleading and must be engineered (split Flats vs Houses).
- New quality features (Flat/House) correlate with Income even stronger; further engineering is recommended for predictive models.
- Future research: explore under which conditions low-income households may own non-garden amenities.

External evidence.

- H1 (inelastic water demand): "Primary uses of water are inelastic; people need water whatever the price. Poor consumers already use the basics, and tariff increases have little effect."
  Source: https://www.pacificwater.org/userfiles/file/IWRM/Toolboxes/financing
- H3 (housing unaffordability in India): "India's housing market is in crisis: home ownership out of reach for middle class. Price-to-Income ratio 8.8 in 2025; market skewed toward luxury housing."
  Source: https://www.fortuneindia.com/business-news/average-home-prices-in-india-are-nearly-9-times-annual-household-income-magicbricks-report/125017

This external evidence on housing unaffordability in India aligns with our finding that Apartment Type and Income Level are strongly correlated.

Final Conclusion

Even with imperfect data, this study demonstrates that systematic exploratory analysis and careful feature engineering can uncover reliable patterns and validate fundamental economic principles. Hypothesis 1 (minimum water demand) and Hypothesis 3 (income–housing link) were confirmed, while Hypothesis 2 was disproved. These findings highlight the importance of data validation and thoughtful preprocessing for any future predictive modeling.